

Time-independent Schrodinger equation

Let $H(x)$ be a time-independent Hamiltonian. Let $\psi(x)$ be a function such that $H(x)\psi(x)$ has eigenvalue E . Hence the time-independent Schrodinger equation

$$H(x)\psi(x) = E\psi(x)$$

Clearly $\psi(x)$ cannot solve the ordinary Schrodinger equation because

$$i\hbar \frac{\partial}{\partial t} \psi(x) = 0$$

Let us construct $\Psi(x, t)$ that is a solution. Recall time-evolution operator $U(t, t_0)$.

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} (t - t_0) H(x) \right)$$

Let $\Psi(x, t)$ be the function

$$\Psi(x, t) = U(t, 0)\psi(x) = \exp \left(-\frac{i}{\hbar} t H(x) \right) \psi(x)$$

Substitute E for $H(x)$.

$$\Psi(x, t) = \exp \left(-\frac{i}{\hbar} Et \right) \psi(x)$$

Now show $\Psi(x, t)$ solves the ordinary Schrodinger equation. By linearity of $H(x)$

$$H(x)\Psi(x, t) = \exp \left(-\frac{i}{\hbar} Et \right) H(x)\psi(x)$$

Substitute E for $H(x)$.

$$H(x)\Psi(x, t) = \exp \left(-\frac{i}{\hbar} Et \right) E\psi(x) \tag{1}$$

Evaluate the Schrodinger time derivative.

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = E \exp \left(-\frac{i}{\hbar} Et \right) \psi(x)$$

Then by equation (1) we have the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = H(x)\Psi(x, t)$$

Hence given the time-independent Schrodinger equation

$$H(x)\psi(x) = E\psi(x)$$

we have

$$\Psi(x, t) = \exp \left(-\frac{i}{\hbar} Et \right) \psi(x)$$

solves the ordinary Schrodinger equation.

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